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Caller ID spoofing effectiveness: A socio-technical analysis of trust propagation and user behaviour

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ABSTRACT

Caller-ID spoofing remains a critical enabler of telephone fraud and social engineering attacks across IP-based and mobile telephony systems. While prior work has analysed signalling vulnerabilities and detection mechanisms, the combined influence of infrastructure-level trust models and human perception remains underexplored. This study presents a socio-technical analysis of caller-ID spoofing by integrating an experimental evaluation of telephony infrastructures with a large-scale survey of user behaviour. We examine spoofing in three representative environments: VoIP systems and SIP trunk interconnections through controlled experiments, and mobile broadband networks using SIM Box devices through a trust-model-based analysis. Results show that spoofing feasibility depends primarily on implicit trust assumptions and validation policies rather than protocol weaknesses alone. Permissive identity validation in VoIP and SIP trunks allows manipulated caller identities to propagate across administrative domains, while SIM Box-based attacks exploit mobile broadband trust models using legitimate mobile identifiers. The user study (332 participants) reveals that decisions are heavily influenced by contextual cues such as institutional identifiers and number familiarity. The findings suggest that the perceived credibility and behavioural impact of spoofed caller identities arise from the interaction of infrastructure trust and human heuristics, highlighting the need for multi-layered mitigation strategies addressing signalling validation, trust relationships, and user awareness. Behavioural findings are based on self-reported intentions rather than observed real-world actions. Statistical results reveal a behavioural separation between call engagement and compliance, where contextual trust cues influence initial interaction, while knowledge and awareness primarily affect subsequent actions.

1. Introduction

Telephone fraud has become a persistent threat in the field of cybersecurity (Şahin et al., 2017), largely driven by the widespread adoption of IP-based telephony infrastructures. The transition from traditional telephone services to Voice over IP (VoIP) technologies has introduced increased flexibility and cost efficiency (Weiss and Kim, 2001), but also new attack vectors related to signalling mechanisms and trust relationships across heterogeneous network elements. In this context, Caller Line Identification (CLI) spoofing has emerged as a key technique for enabling fraud and social engineering attacks such as vishing (Bullée et al., 2016). Unlike legacy telephony systems, modern VoIP and IP-based signalling infrastructures allow call identity parameters to be defined, propagated, or modified across multiple architectural components, including endpoints, signalling servers, switches, and interconnection gateways. This distributed identity management model

facilitates the manipulation of a call origin in the absence of robust authentication and validation mechanisms. The problem extends beyond VoIP environments. Interconnection models based on Session Initiation Protocol (SIP) introduce implicit trust relationships between independent administrative domains that can be exploited to propagate forged identities (Tas and Baktir, 2024). Furthermore, techniques leveraging mobile infrastructures, such as SIM box deployments, enable highly credible spoofing scenarios by operating directly over mobile broadband networks using legitimate numbering resources (Kouam et al., 2021). Although prior work has extensively analysed vulnerabilities in VoIP signalling and proposed detection mechanisms for malicious calls (Rehman and Abbasi, 2014), fewer studies have experimentally evaluated how different telephony infrastructures handle and propagate caller identity when it is manipulated at different points in the system. Moreover, existing mitigation frameworks such as STIR/SHAKEN rely on public key infrastructures that face significant

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deployment and scalability challenges, and are limited to IP-based environments, leaving legacy systems such as SS7 outside their protection scope (Wang et al., 2023). To address the aforementioned limitations, this work presents a combined technical and empirical analysis of caller ID spoofing. Firstly, commonly used spoofing techniques are implemented and evaluated in a controlled experimental environment, including VoIP systems, SIP trunk interconnections, and mobile networks based on SIM box infrastructures. Second, due to the ethical and legal constraints associated with conducting real-world fraud experiments, the technical analysis is complemented by a user study involving 332 participants, aimed at evaluating perception, decision-making, and susceptibility to telephony-based social engineering scenarios. The survey includes SMS-based scenarios to assess user behavioural responses, although technical implementation of SMS spoofing is beyond the scope of this work. Unlike prior work focusing exclusively on signalling vulnerabilities or detection mechanisms, this study jointly examines caller-ID spoofing from both an infrastructure-level and a user-perception perspective, enabling a unified socio-technical analysis.

This paper makes the following contributions:

- A systematic experimental evaluation of caller-ID spoofing across VoIP, SIP trunk, and mobile network environments.
- A scenario-based empirical analysis of user perception and behavioural responses to spoofed calls.
- A socio-technical framework linking infrastructure-level trust propagation and user-level cognitive trust cues.
- Empirical evidence demonstrating a behavioural distinction between call engagement and compliance.

The remainder of this paper is organized as follows. Section II reviews the related work. Section III describes the experimental environment and the methodology employed. Section IV presents the survey and the results obtained across the analysed scenarios, while Section V discusses and compares the findings. The limitations of this study are discussed in Section VI. Section VII concludes the paper and future research directions are outlined in Section VIII.

2. Related work

Caller Line Identification (CLI) spoofing is a well-recognized problem in telephony security, particularly in VoIP-based infrastructures. The flexibility introduced by IP signalling, compared to traditional circuit-switched systems, has expanded the attack surface, and enabled the manipulation of critical signalling parameters, including caller identity (Sherr et al., 2005). Several studies have analysed the inherent vulnerabilities of VoIP signalling and the protocols used for call establishment. In particular, the lack of robust authentication and validation mechanisms in SIP (Ehlert et al., 2010) facilitates abuses such as identity spoofing and service exploitation (Jansky et al., 2017), as well as large-scale abuse of telephony services (Keromytis, 2010). These issues are further exacerbated when SIP servers and private branch exchanges (PBXs) rely on signalling values provided by the origin without applying strict validation policies (Dantu et al., 2009). In this context, it has been observed that the effective security of caller identification largely depends on the trust assumptions between different elements of the infrastructure and on how identity information is propagated along the call path (Gritzalis and Mallios, 2008). This dependency makes CLI an easily exploitable attribute (Mustafa et al., 2016) when signalling is not properly protected (Wang et al., 2026) or when permissive configurations are in place. Regarding unwanted and malicious calls, several detection approaches have been proposed (Li et al., 2018) that avoid relying solely on the declared caller identity. Some works introduce reputation-based models and relationship analysis between users to identify fraudulent calls, demonstrating that behavioural information can be more reliable than CLI when the latter can be manipulated (Azad and Morla, 2013). Other studies have focused on the empirical analysis

of malicious call ecosystems through the observation of real-world attacks against VoIP infrastructures (Prasad et al., 2020). The use of honeypots has enabled the characterization of automated attack patterns and systematic abuse of IP telephony services, highlighting the central role of infrastructure in the propagation of manipulated identities (Carrillo-Mondéjar et al., 2022). Vishing has been studied (Triantafyllopoulos et al., 2025) as an evolution of traditional phishing toward the voice channel, emphasizing its effectiveness when combined with caller ID spoofing techniques. Recent research has proposed early detection mechanisms for telephone fraud on mobile devices, particularly in scenarios where attacks rely on malware or malicious applications (Lee et al., may 2025). Overall, the literature highlights that the feasibility of telephone spoofing is strongly tied to signalling mechanisms and implicit trust models within telephony infrastructures. However, most existing work focuses either on detection mechanisms or on theoretical analyses. There is a lack of experimental studies that systematically evaluate how different infrastructures handle and propagate caller identity when it is manipulated at different points in the system. Furthermore, limited attention has been given to the interaction between technical spoofing capabilities and user perception. This work addresses these gaps by providing a combined experimental and user-centric analysis across VoIP, SIP trunk interconnection scenarios, and mobile network environments.

3. Methodology

3.1. Experimental environment

The experimental evaluation was performed in a controlled telephony environment designed to analyse how caller identity is accepted, modified, validated, or propagated across different infrastructure layers. This section describes the hardware and software components, provider configurations, testing methodology, and ethical safeguards adopted throughout the study.

To facilitate reproducibility and transparency, all experimental artefacts used in this study are publicly available. The repository includes the complete Asterisk and FreePBX configuration files, SIPp scenarios, automation scripts for repeated execution, CSV files containing the results of all experimental repetitions, representative packet captures, anonymized configuration screenshots, and a demonstration video illustrating the evaluated spoofing techniques. The repository also documents the experimental setup, testing methodology, and the ethical and legal constraints under which the experiments were conducted. Together, these materials enable independent verification of the reported results and facilitate replication of the evaluated scenarios. The complete dataset and supporting materials are available at: https://github.com/isifire/Caller_ID_Spoofing_Study_UniOvi.

The experimental environment was based on Asterisk 22.1.1 and FreePBX 17.0.19.23 (Home ★ Asterisk 2026; Download 2026). Calls were originated using the Zoiper SIP client (Zoiper-Free VoIP SIP 2026), which allows manual caller-ID configuration. To ensure consistency and repeatability, each scenario was executed ten times under equivalent conditions. Two commercial SIP trunk providers with different operational policies were evaluated. Provider A offered globally routable SIP services with permissive caller-ID handling, whereas Provider B operated primarily within the Spanish numbering space and imposed routing restrictions on certain numbering ranges. Provider names have been anonymized to avoid publicly identifying commercial vendors whose configurations, if misused, could facilitate caller-ID spoofing.

All experiments were conducted within a clearly defined legal and ethical scope. Only authorized test numbers or arbitrary invented numbers were used, no unauthorized individuals were contacted, and all testing was restricted to infrastructure owned by or explicitly authorized for use by the researchers. SIP trunk experiments were performed using commercial accounts with explicit provider authorization.

The main observation criterion was the comparison between the

identifier configured or manipulated at the call origin and the identifier ultimately presented to the recipient. The objective is to characterize how caller identity is accepted, modified, or propagated under different trust configurations. This criterion permits a direct assessment of whether the identity altered at the origin is accepted, modified, or rejected by the various infrastructure elements, providing an observable and objective indicator of the system's actual behaviour in the face of spoofing attempts. Where possible, signalling exchanged has been analysed to identify where the identity was accepted, modified, or propagated. Signalling events were recorded through server traces and packet captures.

To structure the analysis of spoofing techniques, these were classified according to the point in the infrastructure where the call identifier is manipulated. This approach allows for a systematic comparison of identity behaviour across different telephony environments and an analysis of how assumptions of trust and signalling mechanisms influence each case. The techniques analysed are grouped into: (i) spoofing in VoIP environments, distinguishing between the SIP client, PBX server and SIP protocol levels; (ii) spoofing in interconnection scenarios using SIP trunks; and (iii) spoofing via SIM Box-based infrastructures.

This study focuses on representative VoIP and SIP trunk technologies commonly deployed in European networks (Asterisk/FreePBX, representative of open-source IP-PBX systems; Provider A and Provider B Premium, representative of European commercial trunk providers). Evaluation of other PBX platforms (Cisco CUCM, Avaya, Unify) and worldwide carriers remains for future work. Findings are indicative of implicit-trust-model vulnerabilities but do not claim generality across all VoIP systems or interconnection policies.

3.2. Spoofing in voip environments

3.2.1. SIP client-level spoofing

In this scenario, caller identity manipulation was analysed at the signalling origin, specifically within SIP clients used to initiate VoIP calls. Some SIP clients allow users to manually configure the caller identity included in call signalling messages. When the SIP server does not enforce strict identity validation mechanisms, the identity defined at the client side may be accepted and propagated across the VoIP infrastructure. During the experiments, it was observed that in environments with permissive validation policies, the caller identity configured in the SIP client was presented to the recipient without modification, regardless of whether it corresponded to the authenticated user. Such permissive policies involve validating registration credentials without enforcing consistency between authenticated identities and signalling headers (e.g., From, Contact, P-Asserted-Identity). The absence of caller ID rewriting, lack of restrictions on allowed caller numbers, and acceptance of identifiers not associated with the registered extension further enable this behaviour. Under this implicit trust model, manipulated caller identifiers can be accepted transparently, illustrating the dependence of caller identity integrity on client-server trust relationships.

Across the repeated executions of this scenario, the configured caller identities were consistently accepted and propagated whenever permissive identity-validation policies were in place, indicating stable behaviour under equivalent conditions.

3.2.2. PBX-Level spoofing

A second scenario examined caller identity manipulation at the VoIP PBX level, using a representative software-based IP PBX commonly deployed in enterprise environments. In this case, caller ID alteration does not originate from the SIP client but is instead performed centrally by the PBX, which acts as a call control, routing, and signalling authority within the VoIP architecture. Unlike client-level spoofing, this approach centralizes identity manipulation within a trusted infrastructure component (Buccafurri et al., 2026) that holds routing and signalling privileges. The conducted tests showed that, when administrative access

is available or permissive configurations exist, the PBX can systematically modify the caller identity of outgoing calls (Fig. 1), potentially affecting multiple users simultaneously. From a security perspective, this scenario is particularly critical because the manipulation occurs within a component generally regarded as trusted within the administrative domain. As a result, detection becomes significantly more difficult for mechanisms relying solely on the analysis of the originating SIP client.

The observed caller-ID modifications were reproduced consistently across all repeated executions of the experiment, confirming the stability of the behaviour under the evaluated PBX configuration.

3.2.3. SIP protocol-level spoofing

The final scenario analysed direct manipulation of caller identity at the SIP protocol level, targeting the identity-related fields embedded within SIP signalling messages used during call establishment. These fields are interpreted by network elements to present the call origin to the receiving party. Experimental results demonstrated that, in the absence of adequate controls ensuring signalling integrity and authenticity, the identity values defined within SIP messages can be accepted and propagated throughout the VoIP infrastructure. This behaviour indicates that caller ID security depends heavily on both the validation policies enforced by SIP servers and the protection mechanisms applied to the signalling channel itself. Overall, the evaluated VoIP scenarios show that spoofing feasibility is strongly influenced by where identity manipulation occurs within the infrastructure and by the validation policies enforced by control elements. Manipulation at more trusted or centralized points in the architecture—such as PBXs or signalling paths—generally increases the scope and impact of spoofing while reducing the effectiveness of detection mechanisms.

Identical outcomes were obtained across the repeated SIPp executions, with manipulated identity values being consistently accepted whenever signalling-level validation mechanisms were absent.

To evaluate the influence of defensive identity-validation policies, the SIP protocol-level experiments were repeated under both a permissive Asterisk configuration and the default configuration. Under the permissive configuration, all ten unauthenticated spoofing attempts were accepted, and the manipulated caller identity was propagated unchanged. In contrast, the default configuration consistently rejected all unauthenticated INVITE requests with SIP 401 (Unauthorized) responses across all ten repetitions. These comparative results demonstrate that the observed signalling-level spoofing behaviour is not an inherent property of SIP itself, but rather depends on the authentication and caller-ID validation policies enforced by the receiving infrastructure. The automation scripts, experimental logs, and representative packet captures supporting these observations are included in the public repository to facilitate independent verification and reproducibility.

3.3. Spoofing in SIP trunk interconnection scenarios

This scenario examined caller identity manipulation in interconnection contexts between telephony infrastructures, specifically using SIP trunks. SIP trunking enables calls originating from an IP-based infrastructure to be routed toward external telephony networks, introducing a trust model based on inter-domain and inter-provider agreements. During the evaluation, it was observed that in certain interconnection environments, the caller identity defined at the originating infrastructure was accepted and propagated through the SIP trunk without exhaustive validation of its legitimacy (Fig. 2). This behaviour is particularly relevant, as the implicit trust established between interconnected networks may allow identities that do not correspond to the actual call origin to be presented to the final recipient. Experimental results showed that the configuration of the SIP trunk and the associated interconnection policies directly affect how caller identity is processed. In scenarios with permissive validation rules, the identity configured at the originating infrastructure was transmitted to the

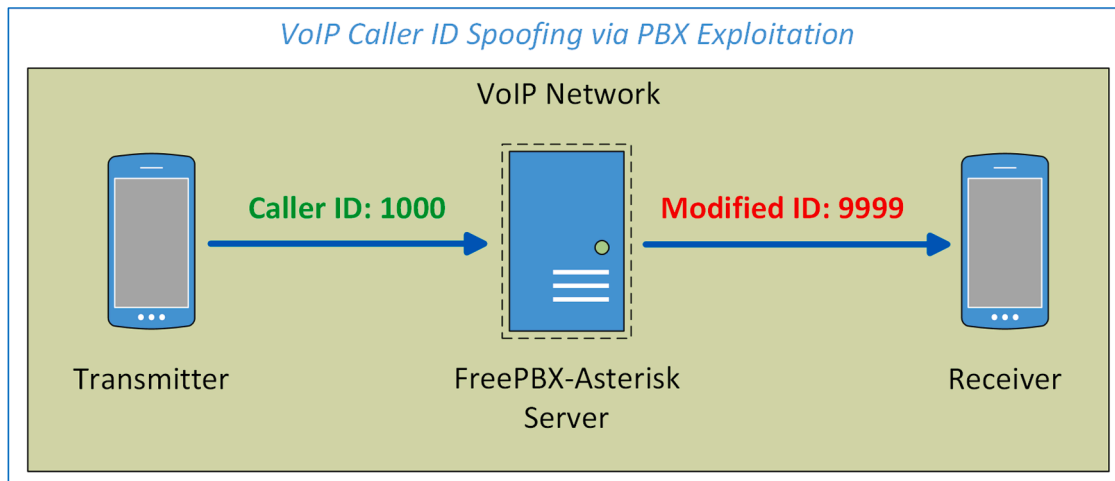


Fig. 1. VoIP caller ID spoofing via PBX exploitation.

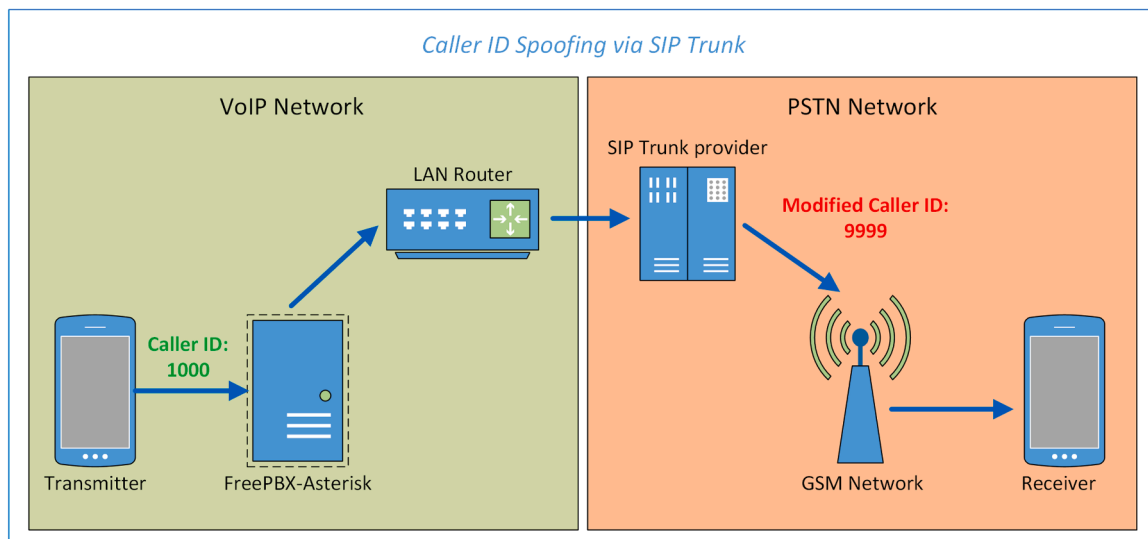


Fig. 2. Caller ID spoofing via SIP trunk.

destination network without modification. This reinforces the role of trunk-level policies in either mitigating or enabling identity spoofing. In addition, these interconnection infrastructures allow the definition of call routing parameters—such as routing rules and charging policies—that facilitate voice traffic management and, in the absence of adequate controls, the propagation of caller identities defined at the source. This capability, while operationally necessary, can unintentionally contribute to identity misuse when validation mechanisms are insufficient.

To evaluate the impact of different interconnection policies, two commercial SIP trunk providers with distinct routing characteristics were analysed. Provider A Premium (Permissive Configuration) accepted caller-ID values in global format without applying geopolitical validation or screening. Across ten repeated spoofing attempts using arbitrary caller identities (e.g., +34 900 001 000, +1 555 123 4567, and other non-assigned test values), all manipulated caller IDs were accepted and propagated to the recipient without modification. Provider B Premium (Operationally Restrictive Routing) was limited to the Spanish numbering space (+34). International caller-ID values were operationally incompatible with the available routes and therefore resulted in rewriting or rejection. Across repeated experiments, international caller-ID attempts (e.g., +1 555 123 4567) were rewritten to the provider's

default Spanish identifier due to route availability constraints rather than explicit caller-ID validation. These results demonstrate that caller-ID propagation is strongly influenced by provider routing and interconnection policies. Provider A enabled arbitrary caller-ID propagation through globally routable services, whereas Provider B operationally restricted certain cross-border scenarios because of route availability. However, these restrictions should not be interpreted as security mechanisms since they do not validate caller identity legitimacy and do not prevent spoofing within the supported numbering space. Scenarios were evaluated in which SIP trunk interconnection was established directly with a VoIP PBX or SIP clients, enabling observation of how caller identities configured in these environments were transmitted across the interconnection infrastructure. The results indicate that SIP trunking introduces an additional cross-domain trust relationship that expands the attack surface for identity spoofing techniques when consistent validation policies are not enforced across interconnected infrastructures.

The complete experimental logs for all repeated SIP trunk tests, together with the corresponding configuration files and supporting documentation, are available in the public repository to facilitate independent verification of the reported observations.

3.4. Spoofing in mobile networks using SIM boxes (Literature-Based analysis)

Unlike the VoIP and SIP trunk scenarios presented previously, this section does not report experimental results obtained by the authors. Instead, it summarizes findings reported in the literature regarding SIM Box-based spoofing and discusses their relevance within the broader caller-ID spoofing landscape.

Telephone spoofing based on SIM Box infrastructure relies on the use of specialized hardware devices capable of hosting multiple SIM cards and connecting directly to the mobile broadband network to originate voice calls or send SMS messages. Unlike spoofing techniques deployed in VoIP environments or SIP trunk interconnection scenarios, this approach operates natively within the mobile network, using valid and active mobile numbers from the perspective of the receiving operator. From the mobile network standpoint, communications generated by a SIM Box are indistinguishable from those originated by legitimate mobile terminals (Oh et al., 2023). The caller identifier presented to the recipient corresponds to a registered and authorized mobile number, which significantly limits the effectiveness of detection mechanisms based on signalling analysis or origin validation at higher layers of the infrastructure (Fig. 3).

The SIM Boxes are specialized devices capable of managing multiple SIM cards and originating mobile traffic. According to the cited literature, SIM Boxes can centralize mobile communications and support automated, high-volume call and SMS traffic. Reported capabilities include caller-ID manipulation, IMEI modification, SIM rotation, and automated traffic generation. Published literature also describes several SIM acquisition strategies that complicate attribution. In some countries, such as the Czech Republic or the United Kingdom (Tskinner, 2026), SIM cards can be obtained without strict identity verification, facilitating their deployment in large-scale fraud campaigns. Other reported practices involve the use of SIM cards registered under third-party identities, further reducing traceability. In Spain, mandatory subscriber identification requirements introduce an additional barrier to domestic SIM Box deployment, although previous studies indicate that this limitation can be partially circumvented through identity misuse or by operating SIM Box infrastructures outside national territory. Taken together, the literature suggests that SIM Box-based spoofing poses challenges that differ from those observed in the VoIP and SIP trunk

scenarios evaluated experimentally in this study. Whereas the experimentally evaluated scenarios exploit trust relationships within IP-based signalling infrastructures, SIM Box operations leverage the trust model of the mobile subscriber ecosystem itself by using legitimate mobile identifiers. As a result, detection mechanisms based solely on caller identity, or signalling-layer validation are likely to be less effective. Direct experimental evaluation of SIM Box infrastructures remains outside the scope of the present work and is identified as an important direction for future research. Table 1 summarizes the technical scenarios discussed throughout this section and highlights the corresponding trust models, validation mechanisms, and operational characteristics.

3.5. Human and perceptual dimension

While the experimental analysis provides a technical perspective on the feasibility and mechanisms of caller-ID spoofing across different telephony infrastructures, it does not capture how such manipulations are perceived by end users nor how they influence real decision-making processes. To complement the infrastructure-level evaluation, a second methodological component was therefore introduced, focusing on the human and perceptual dimension of telephone spoofing. This component adopts a survey-based approach aimed at assessing users' risk perception, trust cues, and behavioural responses when confronted with scenarios representative of the spoofing techniques analysed experimentally. By combining technical experimentation with empirical user data, the methodology enables a more comprehensive evaluation of telephone spoofing, encompassing both its technological feasibility and its impact on end-user vulnerability. The survey instrument was structured into multiple sections designed to capture anonymized demographic attributes, prior cybersecurity-related knowledge, general communication habits and scenario-based behavioural responses. The initial sections gathered anonymized demographic information and self-assessed digital and cybersecurity competencies. Subsequent sections focused on participants' familiarity with relevant fraud-related terminology and the presence or activation of security features on their mobile devices. The core of the questionnaire consisted of a set of controlled scenarios simulating different types of incoming calls—varying the displayed caller identity, number format, or country prefix—and two SMS-based phishing situations. For each scenario, participants were asked to rate the likelihood of answering the

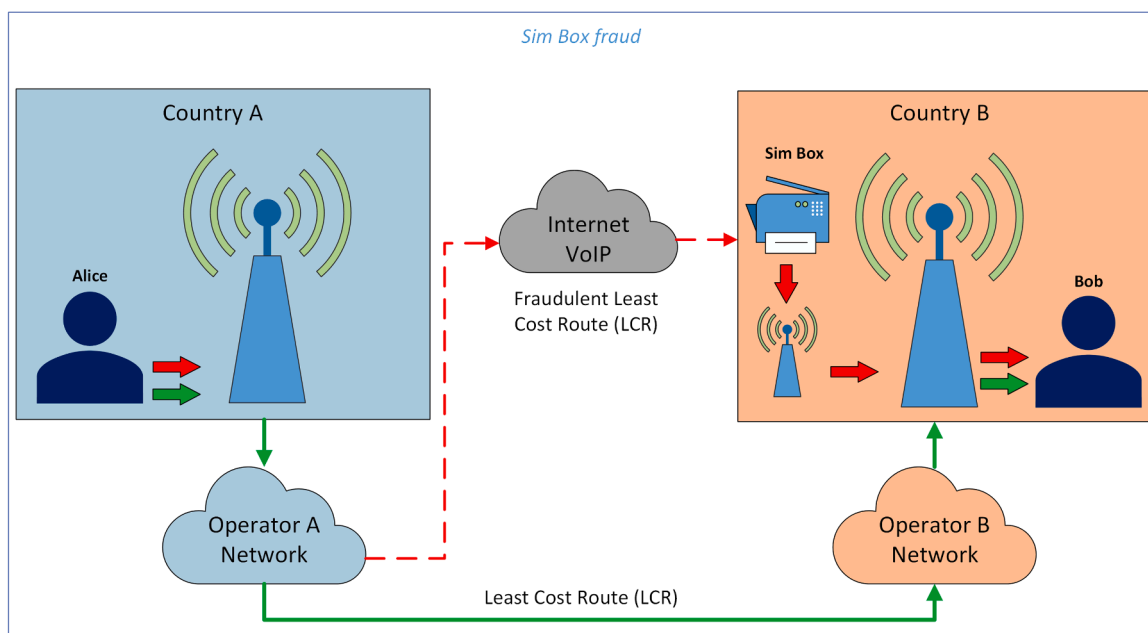


Fig. 3. SIM Box fraud.

Table 1
Summary of technical scenarios.

Aspect	VoIP Environments	SIP Trunk Interconnection	Mobile Networks (SIM Box)
Point of manipulation	SIP client, PBX, or SIP signalling	Originating VoIP/PBX infrastructure or interconnection boundary	Network access via SIM Box
Underlying technology	SIP / VoIP infrastructure	SIP trunk between trusted domains	Native mobile network
Caller-ID origin	Configured at client, PBX, or SIP headers	Defined at originating infrastructure	Legitimate mobile numbers (SIM cards)
Trust model	Implicit trust between SIP client and server	Inter-domain trust between interconnected networks	Trust inherent to mobile subscriber model
Validation controls	Dependent on SIP server and PBX policies	Dependent on trunk configuration and provider policies	Enforced at subscriber registration level
Typical validation weaknesses	Lack of identity consistency checks, no Caller-ID rewriting	Permissive trunk policies, limited cross-domain validation	Use of valid SIMs limits signalling-based detection
Scope of impact	Individual users or internal VoIP domains	Cross-network propagation of spoofed identities	Potentially large-scale, multi-destination campaigns
Detection difficulty	Moderate (depends on server policies)	High (trusted interconnection paths)	Very high (appears as legitimate mobile traffic)
Operational complexity	Low to moderate	Moderate	High (hardware, SIM acquisition, management)
Key security implication	Dependence on SIP trust relationships	Expansion of attack surface across domains	Reliance on legitimate mobile identifiers

communication, perceiving it as legitimate, or following the sender's instructions. The survey concluded with a general question assessing the perceived feasibility of caller-ID spoofing. This structure ensured a progressive transition from baseline characteristics to context-dependent behavioural assessments, enabling consistent comparison across scenarios. The survey design follows a scenario-based (vignette) methodology, widely used in human factors and security research to evaluate behavioural responses under controlled yet realistic conditions. By presenting participants with standardized hypothetical situations, this approach enables consistent comparison of user decisions across varying contextual factors while avoiding ethical and legal concerns associated with real-world deception experiments. All behavioural responses were measured using five-point Likert scales ranging from "Very improbable" to "Very probable". Likert scales were selected due to their suitability for capturing subjective assessments of probability, trust, and intended behaviour in a structured and comparable manner. Given the ordinal nature of these measures, subsequent statistical analysis relied on nonparametric methods that do not assume normal distribution of responses. This methodological approach ensures that observed behavioural patterns can be systematically analysed while maintaining both experimental control and ecological validity in the evaluation of spoofing-related decision-making. This approach enables the isolation of contextual trust cues associated with caller identity presentation, allowing direct comparison with the technical spoofing scenarios analysed in the experimental component of this study.

4. Results

The study sample consisted of 332 participants, recruited through open dissemination on social networks and institutional channels at the University of Oviedo (shown at Table 2). The sample shows a balanced demographic distribution, with the 18–29 age group representing the largest segment (38.86%), followed by participants aged 45–59

Table 2
Sociodemographic characteristics of the sample.

Variable	Category	Frequency (n)	Percentage (%)
Age group	18–29 years	129	38.86
	45–59 years	102	30.72
	> 60 years	51	15.36
	30–44 years	50	15.06
Gender	Male	174	52.41
	Female	153	46.08
	Prefer not to say	5	1.51
Education level	University degree	132	39.76
	Postgraduate degree	128	38.55
	Secondary education	61	18.37
	Basic education	11	3.31

(30.72%), ≥ 60 (15.36%), and 30–44 (15.06%). Gender distribution was nearly balanced (52.41% male, 46.08% female), and the overall educational level was high, with 78.31% holding a university or post-graduate degree. Participants reported predominantly medium-to-high levels of digital skills and baseline to medium self-perceived cybersecurity knowledge (shown at Table 3), providing a suitable context for analysing behavioural responses to spoofing and fraudulent communication scenarios.

Table 4 summarizes participants' prior knowledge related to telephone fraud and the configuration of security features on their mobile devices. These variables provide important contextual information for interpreting individual susceptibility to spoofing and smishing attempts, as they relate to users' familiarity with key cybersecurity concepts and the protective measures already enabled on their smartphones. Overall, the data reveal limited awareness of technical terminology but widespread adoption, often unintentional, of built-in anti-fraud features, which is relevant when assessing behavioural responses throughout the experimental scenarios.

Table 5 presents participants' overall perception of the feasibility of telephone number spoofing. A majority of respondents (57.2%) considered spoofing to be feasible, while 9.0% perceived it as difficult or impossible. A substantial proportion (33.7%) expressed uncertainty. Notably, despite the majority perceiving spoofing as feasible, a large fraction of participants either underestimate its feasibility or lack a clear understanding of it. This dispersion suggests the absence of a consistent mental model regarding caller-ID manipulation, which is particularly relevant when interpreting behavioural responses in later scenarios. In this context, uncertainty and misperception may contribute to reliance on heuristic trust cues rather than informed risk assessment.

This variability in perceived risk constitutes an important contextual factor for subsequent analyses, as it reflects differing initial assumptions

Table 3
Self-perceived digital and cybersecurity skills.

Variable	Category	Frequency (n)	Percentage (%)
Self-perceived digital skills	Medium	143	43.07
	High	120	36.14
	Low	38	11.45
	Very high	29	8.73
	Very low	2	0.60
Self-perceived cybersecurity knowledge	Medium	120	36.14
	Basic	109	32.83
	High	67	20.18
	Very low / none	30	9.04
	Expert	6	1.81

Table 4
Prior knowledge.

Variable	Category	Frequency (n)	Percentage (%)
Use of online banking	Yes	294	88.55
	No	38	11.45
Knowledge of the term “telephone spoofing”	No	210	63.25
	Yes	122	36.75
Knowledge of the term “smishing”	No	252	75.90
	Yes	80	24.10
Phone includes anti-fraud / call-blocking features	Yes	225	67.77
	Don’t know	85	25.60
	No	22	6.63
Anti-fraud features currently activated	Yes	239	71.99
	No	55	16.57
	Don’t know	38	11.45

Table 5
Participants’ perception of caller-ID spoofing feasibility (N = 332).

Perception	Frequency (n)	Percentage (%)
Feasible (easy or relatively easy)	190	57.2
Difficult or not feasible	30	9.0
Uncertain / Do not know	112	33.7

about the plausibility of fraudulent communications and may influence participants’ likelihood of engaging with or dismissing the simulated call scenarios. The experimental section of the survey comprised five call-based scenarios, as shown at Table 6, each designed to replicate common conditions under which fraudulent communications may occur. The call scenarios varied the displayed caller identity, including the exact name of the participant’s bank, an official customer-service number, an unknown national number, an international prefix, and a

Table 6
Comparative likert scale responses across call scenarios.

Scenario	Displayed Caller Information	Response Dimensions (Likert 1–5)	Most Frequent Response Categories
Scenario 1	Exact bank name shown on screen	Likelihood of answering; likelihood of perceiving legitimacy; likelihood of following instructions	“Very improbable” or “Poorly probable” dominate follow-instruction responses; mixed distribution for answering and legitimacy.
Scenario 2	Official customer-service number	Same three Likert responses	Higher proportions of “Probable/Neutral” for answering and legitimacy compared to other scenarios.
Scenario 3	Unknown number	Same three Likert responses	“Very improbable” most frequent across all three dimensions.
Scenario 4	International prefix	Same three Likert responses	Strong predominance of “Very improbable” for answering, legitimacy, and following instructions.
Scenario 5	Number similar to participant’s own	Same three Likert responses	Slightly more variability, with increases in “Probable” and “Neutral” for answering compared to Scenarios 3–4. “Very improbable” remains dominant for following instructions.

number visually like the participant’s own. For each of these conditions, respondents rated (i) the likelihood of answering the call, (ii) the likelihood of perceiving it as legitimate, and (iii) the likelihood of following the caller’s instructions. All ratings were collected using a five-point Likert scale, ranging from “Very improbable” to “Very probable”. Two additional SMS-based scenarios replicating smishing attempts were studied, prompting respondents to evaluate message legitimacy, their likelihood of calling back or clicking on embedded links, and their ability to identify suspicious elements using categorical Likert-type responses; however, these scenarios are beyond the scope of the present work. This design allowed consistent comparative analysis across scenarios and response dimensions.

Several relevant patterns emerge from the comparison of call-based scenarios (Fig. 4). First, scenarios displaying credible or familiar caller information—such as the exact bank name or an official customer-service number—show a broader distribution of likelihood ratings for answering and for perceiving the call as legitimate, with higher proportions of “Probable” and “Neutral” selections relative to the remaining scenarios. Second, scenarios involving ambiguous or non-familiar caller IDs, namely unknown numbers, international prefixes, or artificially similar numbers, consistently exhibit a predominance of “Very improbable” responses across the three behavioural dimensions. Finally, across all scenarios, the likelihood of following caller instructions is systematically lower than the likelihood of answering or perceiving legitimacy, with “Very improbable” being the modal response in every condition. These descriptive findings provide a structured basis for the subsequent analyses examining behavioral sensitivity to spoofing indicators.

Taken together, these aggregated results provide a consolidated view of how participants responded across all call-based scenarios and behavioural dimensions. The combined distribution highlights clear differences in response patterns between the likelihood of answering or perceiving legitimacy and the likelihood of following instructions, with the latter consistently showing the strongest bias toward low-probability categories. This integrated representation serves as a foundation for the subsequent Discussion section, where these patterns can be examined in relation to participants’ prior knowledge, perceived risk, and the specific contextual cues embedded within each scenario. The following discussion contextualizes these findings within broader human-factors considerations and explores their implications for user susceptibility to caller-ID spoofing and related fraud attempts.

As revealed at Table 7, the survey results show a significant lack of user awareness regarding telephone numbering schemes. For a legitimate Spanish mobile number, only 42.5% of respondents correctly identified it, while more than 50% expressed uncertainty or misclassified it as fraudulent. Similarly, although international numbers were more frequently recognized (53.3%), nearly half of the participants either misidentified them or reported uncertainty. These findings highlight that users rely on simplistic heuristics and exhibit limited confidence when interpreting caller ID information. This cognitive gap can be effectively exploited in caller ID spoofing attacks, where attackers manipulate perceived trust by mimicking familiar or ambiguous number formats.

4.1. Statistical analysis

To complement the descriptive findings, a comprehensive set of nonparametric statistical analyses was conducted to evaluate the influence of scenario characteristics, user knowledge, technical protections, and demographic variables on behavioural responses. All behavioural variables were measured using five-point Likert scales and treated as ordinal data. Consequently, nonparametric statistical methods were employed throughout. All tests were two-tailed with a significance level of $\alpha = 0.05$.

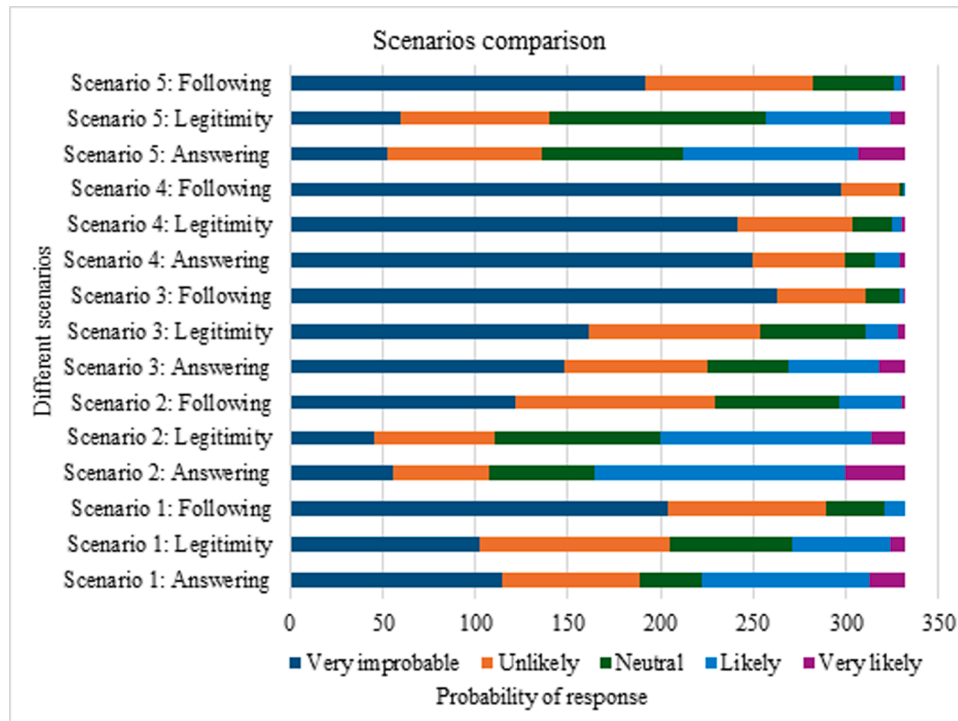


Fig. 4. Scenarios comparison: a) Scenario 1: bank name on screen b) Scenario 2: bank's official number c) Scenario 3: unknown number d) Scenario 4: International number, e) Scenario 5: similar number (same prefix).

Table 7
Caller identification capabilities.

Answer	Number of answers (n)	
	Call from +34 886 49 95 97 (Spanish mobile phone number)	Call from +1 855 330 8653 (International call)
Spanish mobile phone number	141	1
I don't know	115	77
It is a fraudulent number as it does not exist	66	77
International call	10	177

4.1.1. Scenario-Level effects (Primary analysis)

A Friedman test was conducted to assess whether behavioural responses differed across the five call scenarios for each dimension. The analysis revealed highly significant differences across scenarios for all behavioural measures:

- Likelihood of answering: $\chi^2(4) = 481.18$, $p < 0.001$
- Perceived legitimacy: $\chi^2(4) = 517.22$, $p < 0.001$
- Likelihood of following instructions: $\chi^2(4) = 390.23$, $p < 0.001$

These results demonstrate that user behaviour is strongly scenario-dependent, confirming that contextual factors such as caller identity, numbering format, and perceived institutional legitimacy significantly influence decision-making.

4.1.2. Cybersecurity knowledge and behaviour

Spearman's rank correlation analyses show that self-perceived cybersecurity knowledge is not significantly associated with likelihood of answering calls, or perceived legitimacy across any scenario. However, statistically significant negative correlations were found between cybersecurity knowledge and the likelihood of following instructions in

multiple scenarios:

- Scenario 1: $\rho = -0.113$, $p = 0.039$
- Scenario 3: $\rho = -0.147$, $p = 0.007$
- Scenario 5: $\rho = -0.117$, $p = 0.033$

4.1.3. Awareness of spoofing and behaviour

Statistically significant differences were observed in the likelihood of following instructions across multiple scenarios (Scenarios 1, 2, 3, and 5), indicating that participants aware of spoofing concepts consistently exhibited lower willingness to comply with caller instructions.

4.1.4. Impact of anti-fraud features

The Mann-Whitney U test indicates only limited and scenario-specific effects, with significant differences in some cases (e.g., Scenarios 1 and 3), but no consistent pattern across behavioural dimensions. No significant differences were observed for perceived legitimacy. These results suggest that technical protections alone do not consistently influence user perception or behaviour, particularly in high-trust scenarios.

4.1.5. Demographic control variables

Spearman correlation analysis shows that demographic effects are statistically weak:

- Age is negatively correlated with answering behaviour ($\rho = -0.109$, $p = 0.047$)

Education level is positively correlated with:

- answering ($\rho = 0.215$, $p < 0.001$)
- perceived legitimacy ($\rho = 0.183$, $p < 0.001$)
- following instructions ($\rho = 0.110$, $p = 0.046$)

However, effect sizes remain modest, indicating that demographic variables do not account for the strong scenario-level behavioural

differences.

4.1.6. Key behavioural pattern: engagement vs. compliance

Across all analyses, a consistent and statistically supported pattern emerges: factors such as scenario context, cybersecurity knowledge, and spoofing awareness do not significantly affect the likelihood of answering calls or perceiving legitimacy. However, these same factors do significantly reduce the likelihood of following caller instructions. This demonstrates a clear behavioural distinction between:

- initial engagement (answering / perceived legitimacy)
- subsequent compliance (following instructions)

This separation suggests that users rely on heuristic trust cues to engage with calls, while applying additional cognitive evaluation before executing potentially risky actions. Detailed statistical outputs for all analyses are provided in the Appendix.

5. Discussion

5.1. Trust propagation across technical and human layers

The experimental results presented in this work provide complementary perspectives on caller-ID spoofing, enabling a joint examination of its technical feasibility and its perceived credibility from the user perspective. While the experimental analysis illustrates how caller identity can be manipulated and propagated across different telephony infrastructures depending on trust models and validation policies, the survey results offer insight into how end users interpret and react to these identities in realistic interaction scenarios. In particular, the results suggest that the same trust relationships that enable identity propagation within VoIP, SIP interconnection, or mobile networks are also reflected in users' behavioural responses, influencing their willingness to answer calls, perceive legitimacy, or follow instructions. This section discusses the implications of these observations by integrating the technical experimentation with the empirical user data. First, it examines how infrastructure-level trust models align with user perceptions of caller identity. It then explores behavioural patterns and statistically significant differences across scenarios, highlighting how contextual cues and prior knowledge affect user decision-making. Together, these perspectives provide a socio-technical explanation of caller-ID spoofing effectiveness by linking infrastructure-level trust assumptions with user-level trust judgments.

The technical analysis reveals that caller-ID spoofing feasibility is primarily governed by the trust assumptions embedded within each telephony infrastructure. Across VoIP environments, SIP trunk interconnections, and mobile networks, the acceptance and propagation of caller identity depend less on the protocol itself than on where identity is validated, and which entities are implicitly trusted. In VoIP environments, trust is typically localized within an administratively controlled domain, where SIP servers and PBXs act as authoritative intermediaries. When validation policies are permissive, identity values defined at the client, PBX, or signalling level are accepted without reconciliation against the authenticated user. The experimental results demonstrate that such configurations enable caller-ID manipulation at multiple points in the infrastructure. From the user perspective, the survey data suggest that identifiers commonly associated with trusted institutions (such as bank names displayed on screen) are more likely to be interpreted as legitimate, even when no technical guarantees exist. This alignment indicates that technical trust delegation at the infrastructure level is mirrored by cognitive trust delegation at the user level, increasing the likelihood that spoofed identities are perceived as credible by users in these environments. SIP trunk interconnection scenarios introduce a broader trust boundary (Strand and Leister, 2011), extending beyond a single administrative domain. Here, caller identity acceptance is influenced by inter-provider agreements and trunk-level

configurations rather than end-user authentication. The experimental findings show that, under permissive interconnection policies, identities supplied by the originating domain may be propagated across networks without modification. Correspondingly, survey responses indicate increased user confidence when calls appear to originate from official or externally validated sources, such as well-known customer-service numbers. This interpretation is further supported by the statistically significant scenario-level differences observed across all behavioural dimensions, indicating that caller identity presentation has a measurable impact on user responses. This suggests that cross-domain trust relationships not only expand the technical surface for identity misuse but also reinforce users' perception that such calls are inherently more credible, despite the absence of end-to-end identity verification. In mobile network scenarios based on SIM Box infrastructures, trust is derived from the mobile subscriber model itself. Calls originate using legitimate, registered mobile numbers that are indistinguishable from those of regular users from the network's perspective. As shown by the experimental analysis, this model significantly limits the effectiveness of signalling-level detection mechanisms. Survey results further indicate that users tend to attribute greater legitimacy to caller identifiers that appear consistent with familiar national numbering conventions, particularly when uncertainty regarding the caller's identity is high. This convergence underscores that the inherent trust associated with valid mobile identifiers translates directly into perceived authenticity, making SIM Box-based spoofing especially challenging to detect both technically and cognitively. Taken together, these observations suggest that caller-ID spoofing exploits a shared trust continuum spanning infrastructure components and human decision-making. The observed separation between engagement and compliance provides additional support for this socio-technical interpretation. While infrastructure-level trust models enable the propagation of credible caller identities that facilitate user engagement, individual knowledge and awareness primarily influence behaviour at the point of action. This asymmetry highlights that technical and cognitive trust mechanisms operate at different stages of the interaction, which has direct implications for both attack design and defence strategies. Technical trust models (based on administrative control, inter-domain agreements, or subscriber legitimacy) define how identities are propagated, while users rely on corresponding heuristics to assess legitimacy. The resulting relationship between system-level trust and user perception helps to explain why forged identities that are technically valid often succeed in capturing the user's attention and provides a conceptual basis for analysing behavioural patterns and mitigation challenges.

From this perspective, caller-ID spoofing can be understood as the propagation of trust across successive technical and cognitive layers rather than as an isolated signalling weakness. Technical infrastructures determine whether a manipulated identity is accepted and propagated, whereas users evaluate that propagated identity using contextual heuristics such as institutional affiliation, numbering conventions, or familiarity. The effectiveness of caller-ID spoofing therefore emerges from the interaction between these two layers: infrastructure-level trust creates the conditions for credible identity presentation, while cognitive trust determines whether that identity influences user behaviour. This socio-technical interpretation provides a unifying framework for understanding why improvements restricted to either signalling security or user awareness alone are unlikely to fully mitigate caller-ID spoofing attacks.

5.2. Behavioural patterns: engagement vs compliance

The scenario-based survey results reveal consistent behavioural patterns (Akinyemi et al., 2022) that align with the trust models identified in the technical analysis. These behavioural differences are statistically corroborated by scenario-level analysis. A Friedman test revealed large scenario-level effects (Kendall's $W = 0.29\text{--}0.39$, $p < 0.001$) across all five call scenarios for the likelihood of answering calls,

perceived legitimacy, and willingness to follow instructions. These findings were further supported by post-hoc pairwise comparisons, which confirmed statistically significant differences across nearly all scenario pairs (Appendix D). This result demonstrates that user responses are strongly scenario-dependent and cannot be explained solely by individual characteristics, reinforcing the central role of contextual trust cues in shaping behavioural responses to caller-ID spoofing. Across all call scenarios, participants' responses vary systematically depending on the contextual cues associated with the displayed caller identity, suggesting that users rely on heuristic trust signals rather than explicit verification when assessing incoming calls (Benenson et al., 2017). Scenarios presenting familiar or institutionally associated identifiers, such as the explicit bank name or an official customer-service number, elicited a wider dispersion of responses across the Likert scale. In these cases, participants showed increased willingness to answer the call and to perceive it as legitimate when compared to scenarios involving unknown or ambiguous identifiers. This behaviour indicates that the presence of recognizable organizational cues can lower users' initial scepticism, even in the absence of technical guarantees regarding caller authenticity. Notably, this effect mirrors the experimental results observed in VoIP and SIP trunk environments, where trusted identifiers are propagated based on permissive validation policies. Scenarios involving unknown numbers or international prefixes produced a pronounced shift toward defensive behaviour. For these scenarios, "Very improbable" consistently emerged as the dominant response category across all three behavioural dimensions—answering, perceived legitimacy, and following instructions. This pattern suggests that numerical unfamiliarity acts as a strong deterrent, reinforcing user caution when trust signals are weak or absent. From a behavioural standpoint, this response reflects a risk-averse heuristic commonly employed when the perceived cost of engagement outweighs the expected benefit. The scenario presenting a number visually like the participant's own generated a more nuanced behavioural profile. While caution remained prevalent—particularly regarding the likelihood of following instructions—responses showed greater variability in answering behaviour. This suggests that ambiguous familiarity can partially override initial scepticism, creating a borderline trust condition in which users neither fully reject nor confidently accept the call. Such ambiguity is particularly relevant for spoofing techniques that deliberately exploit subtle numerical similarities to induce momentary hesitation or confusion. Across all scenarios, one pattern remained consistent: participants were significantly less willing to follow caller instructions than to answer the call or perceive it as legitimate. This behavioural separation between initial engagement and subsequent compliance is further reinforced by the statistical analysis. While scenario-level factors exert the strongest influence on the likelihood of answering calls and perceived legitimacy, cybersecurity knowledge and spoofing awareness show limited or non-significant effects on these initial responses. In contrast, both variables are consistently associated with a lower willingness to follow caller instructions. Even in scenarios associated with higher perceived legitimacy, following instructions displayed a strong bias toward low-probability categories. This separation between initial engagement and compliance highlights an important behavioural threshold, indicating that users apply additional cognitive evaluation before committing to potentially risky actions. From a security perspective, this behaviour suggests that successful spoofing does not necessarily require immediate compliance, but rather the establishment of sufficient trust to prolong interaction. Overall, these behavioural patterns demonstrate that user responses to spoofed calls are highly scenario-dependent and context-driven, closely reflecting the implicit trust cues leveraged by different spoofing techniques. The findings underscore that behavioural responses to caller-ID spoofing are shaped not only by technical feasibility but also by how users interpret and prioritize identity signals under uncertainty. These observations provide the qualitative foundation for the statistical analyses presented in the following section, where the significance of behavioural differences

across scenarios and user characteristics is formally evaluated.

Several survey results related to prior knowledge and self-assessment provide relevant context for interpreting user behaviour in caller-ID spoofing scenarios. These variables do not only describe the sample, but also help explain the observed differences in behavioural responses across scenarios.

5.2.1. Low conceptual awareness despite high exposure to digital services

Although a large majority of participants report regular use of online banking services (88.55%), direct awareness of telephone fraud terminology remains limited. Only 36.75% of respondents indicated familiarity with the term telephone spoofing, and recognition of smishing was even lower, at 24.10%. Considering that online banking is a common target in both voice- and message-based fraud, this discrepancy highlights a clear mismatch between risk exposure and conceptual understanding of the underlying attack techniques. From a behavioural perspective, this gap suggests that many users may encounter spoofing attempts without possessing the vocabulary or mental models required to explicitly identify them.

5.2.2. High prevalence of built-in security features often activated unknowingly

A substantial proportion of participants reported that their mobile phones include anti-fraud or call-blocking features (67.77%), and an even higher percentage stated that such protections were currently activated (71.99%). When contrasted with the limited awareness of spoofing terminology, these figures indicate that a significant fraction of protection is likely provided by default platform or operator mechanisms, rather than by deliberate user configuration. This observation aligns with the increasing role of operating systems and carriers in deploying automated filtering and detection mechanisms, partially compensating for limited user knowledge.

5.2.3. Uncertainty in security self-assessment reflected in "Don't know" responses

Despite the widespread availability of anti-fraud features, uncertainty remains pronounced. Approximately one quarter of respondents (25.60%) were unsure whether their device supported such capabilities, and 11.45% did not know whether these features were currently enabled. This uncertainty suggests that users may have difficulty accurately assessing their own security posture, potentially leading to over- or under-estimation of protection. In ambiguous call scenarios, such uncertainty may translate into inconsistent behaviour, as users cannot reliably judge whether technical safeguards are actively reducing risk.

5.2.4. A risk-significant user profile emerges from combined indicators

The combination of frequent engagement in high-value digital services (such as online banking), limited conceptual awareness of telephony fraud, and partial uncertainty regarding protective mechanisms defines a risk-relevant user profile. Users within this profile are routinely exposed to spoofing threats but may rely primarily on heuristic cues—such as caller identity presentation—rather than informed judgment when assessing legitimacy. This context is particularly relevant for interpreting the behavioural results observed in call-based scenarios, where technically credible spoofed identities elicited greater engagement despite the absence of verifiable authentication. Collectively, these findings indicate that susceptibility to telephone spoofing cannot be explained solely by lack of knowledge or by the absence of technical protections. Instead, it emerges from the interaction between implicit trust in infrastructure-level defences, limited explicit understanding of fraud mechanisms, and uncertainty in personal security assessment. This interaction provides a meaningful explanatory framework for the scenario-dependent behavioural patterns analysed in the preceding sections.

5.2.5. Statistical relationships between knowledge, protection measures and behavioural responses

To further support the qualitative interpretation, statistical analyses reveal consistent relationships between participants' prior knowledge, self-assessment, and their behavioural responses in spoofing scenarios. Correlation analyses show that higher self-perceived cybersecurity knowledge does not significantly reduce the likelihood of answering spoofed calls nor the perceived legitimacy of the caller. However, cybersecurity knowledge is consistently associated with a statistically significant reduction in the willingness to follow caller instructions in several scenarios. This distinction indicates that security knowledge primarily influences behaviour at the point of action rather than during initial engagement, separating exposure from compliance. A similar pattern is observed for conceptual awareness: familiarity with the term "telephone spoofing" does not significantly affect answering behaviour or legitimacy perception but is associated with a lower willingness to comply with caller instructions across most scenarios. Group-based comparisons indicate that participants who reported having anti-fraud or call-blocking features activated on their devices exhibited more conservative behaviour in a limited number of scenarios, particularly with respect to answering calls and following instructions. However, these effects were not consistently observed across all caller-identity conditions. This suggests that technical protections alone do not eliminate behavioural risk, but rather interact with user perception, trust assessment, and scenario context. Finally, scenario-level analyses confirm statistically significant differences in behavioural responses across caller identity conditions, supporting the conclusion that contextual trust cues play a decisive role in shaping user decisions. Taken together, these statistical associations reinforce the view that behavioural responses to telephone spoofing emerge from the interaction between prior knowledge, perceived protection, and scenario-specific trust signals, rather than from any single factor in isolation.

Taken as a whole, the discussion underscores that caller-ID spoofing is best understood as a socio-technical phenomenon emerging from the interaction between infrastructure-level trust delegation and human decision-making heuristics. The experimental analysis demonstrates that identity manipulation is feasible across multiple telephony environments, with its success largely defined by where identity validation occurs, and which entities are implicitly trusted. The survey results complement this view by showing that users respond to these trust signals in systematic ways, relying on contextual cues rather than explicit verification to assess legitimacy. Across VoIP, SIP trunk, and mobile network scenarios, the alignment between technical trust models and user perception creates conditions under which spoofed identities can appear both technically valid and cognitively credible. Statistical analyses further indicate that the influence of these factors varies across caller identities and trust contexts, reinforcing the importance of scenario-specific cues in user decision-making. These findings help explain why caller-ID spoofing can remain behaviourally persuasive despite increased awareness of telephony fraud and the deployment of technical countermeasures.

5.3. Defensive configurations and mitigation

The experimental results demonstrate that caller-ID spoofing feasibility is strongly influenced by validation policies and trust relationships within telephony infrastructures. Consequently, the findings have direct implications for the design of mitigation mechanisms at the VoIP, interconnection, and carrier levels. This section discusses the defensive configurations evaluated in the study and situates them within the broader context of telephony security.

5.3.1. VoIP-Level defences: asterisk default configuration

Authentication and Identity Validation.

The evaluation of Asterisk's restrictive configuration identified three complementary mechanisms that prevented client-level spoofing in the

evaluated scenarios: SIP credential authentication, From-header validation, and caller-ID rewriting.

When SIP authentication was enforced, all spoofing attempts initiated without valid credentials were rejected before reaching the dial plan. Similarly, From-header validation prevented identity impersonation by ensuring consistency between the authenticated user and the presented caller identity. Across all ten repetitions, spoofed From-header values were rejected successfully.

A third mechanism, caller-ID rewriting, forced CALLERID(num) to match the authenticated extension regardless of the value supplied by the client. In all evaluated scenarios, manipulated caller identities were overwritten before reaching the recipient endpoint.

Taken together, these observations illustrate that spoofing feasibility in the evaluated VoIP environments is determined largely by infrastructure trust policies rather than by protocol limitations alone. Under restrictive configurations, all evaluated spoofing attempts were either blocked or rendered ineffective, whereas equivalent attacks succeeded under permissive configurations.

5.3.2. Endpoint-Level illustration

To further illustrate the practical impact of caller-ID spoofing at the telephony endpoint level, a demonstration video (Video S1, available in the supplementary repository) was recorded using an analogue rotary telephone connected via an Analogue Telephone Adapter (ATA) to the Asterisk PBX. In the first segment, a legitimate call is placed from Phone 1 to Phone 2, which correctly displays Phone 1's stored contact name and number, establishing the expected baseline behaviour. In the second segment, a call is placed from the analogue rotary telephone to Phone 2 with the Caller ID spoofed to match Phone 1's number. Despite originating from a physically distinct device, Phone 2 displays the same contact name and number as in the legitimate call, demonstrating that the recipient cannot distinguish the spoofed call from a genuine one based on the caller identification alone. This illustrates, at the endpoint level, the same implicit-trust mechanism discussed throughout [Section 3](#): caller identification is accepted and displayed without independent verification of the originating device or subscriber.

5.3.3. Carrier-Level authentication: stir/shaken

STIR/SHAKEN represents one of the principal standards-based approaches for mitigating caller-ID spoofing through digital signatures and identity attestation. In principle, full attestation mechanisms can provide a strong level of assurance regarding caller identity legitimacy. However, practical effectiveness remains dependent on widespread deployment across all operators involved in the call path.

In many jurisdictions, including large parts of Europe, deployment remains incomplete and implementation policies vary across carriers. As a result, spoofing opportunities continue to exist in heterogeneous environments where authentication coverage is inconsistent.

Because the present study focuses on environments where STIR/SHAKEN is not universally deployed, direct evaluation of its effectiveness remains beyond the scope of this work. Nevertheless, broader adoption of carrier-level authentication frameworks represents an important direction for future mitigation efforts and warrants further empirical evaluation under heterogeneous deployment conditions.

5.3.4. Interconnection-Level limitations

The comparative analysis of the evaluated SIP trunk providers highlights an important distinction between operational restrictions and security controls.

Although routing constraints limited certain international spoofing scenarios in Provider B (see section 3.3), these restrictions did not perform caller-identity legitimacy checks and therefore cannot be considered security mechanisms. Domestic caller-ID manipulation remained feasible whenever the supplied identity belonged to the supported numbering space.

These observations indicate that routing policies alone are

insufficient to prevent spoofing. Effective protection requires explicit caller-identity validation and authentication mechanisms rather than reliance on operational limitations imposed by routing scope.

Taken together, the evaluated defensive mechanisms highlight substantial differences in effectiveness across architectural layers. While authentication, identity-validation, and caller-ID consistency enforcement effectively prevented spoofing in the evaluated VoIP environments, operational routing constraints alone do not provide meaningful protection against caller-ID manipulation. Standards-based approaches such as STIR/SHAKEN offer a promising long-term solution, although their practical effectiveness currently depends on deployment coverage and enforcement policies. A comparative summary of the evaluated mitigation mechanisms is presented in Table 8.

6. Limitations

While the results presented in this study provide a comprehensive view of caller-ID spoofing from both technical and user-behavioural perspectives, several limitations should be acknowledged when interpreting the findings.

First, the experimental analysis was conducted in controlled environments designed to isolate specific spoofing techniques and trust configurations. Although the evaluated VoIP, SIP trunk, and mobile network scenarios reflect commonly deployed infrastructures, real-world deployments may exhibit additional complexity due to heterogeneous configurations, operator-specific policies, and evolving interconnection agreements. Consequently, the results should be interpreted as indicative of structural behaviours and trust dependencies, rather than as an exhaustive characterization of all operational telephony systems.

Second, ethical, and legal constraints precluded the execution of real-world spoofing attacks against unsuspecting users. As a result, the experimental component focused on observing identity propagation and validation behaviour within infrastructure elements, while the human impact was assessed through simulated scenarios in a controlled survey setting. Although this approach avoids exposing participants to actual harm, it may not capture all contextual pressures present during real attacks, such as emotional manipulation or time-critical decision-making.

Third, the survey sample, while sizeable and diverse, is not intended to be fully representative of the general population. Participants were recruited primarily through academic and online channels, resulting in a sample with relatively high levels of education and digital engagement. This characteristic may influence baseline awareness and behavioural tendencies. However, this limitation also implies that the observed susceptibility patterns may be conservative, as even a digitally experienced population exhibited uncertainty and risk-prone behaviour under specific spoofing conditions.

Fourth, behavioural responses were collected using self-reported

Likert-scale measures, which capture stated intention rather than actual behaviour. Although this method is widely used in security and human-factors research, it cannot fully account for discrepancies between reported willingness and real-world actions. Nonetheless, the consistency of patterns across scenarios and their alignment with infrastructure-level trust models suggests that the results capture meaningful behavioural tendencies.

Finally, the statistical analyses presented focus on scenario-level effects and associations between knowledge, self-assessment, and behaviour. While these analyses support the socio-technical interpretation of spoofing risk, they do not establish causal relationships. Future work could extend these insights through longitudinal studies, controlled experiments, or field deployments involving adaptive spoofing or real-time user feedback mechanisms.

7. Conclusions

This work has presented a comprehensive analysis of caller-ID spoofing by integrating experimental evaluation of telephony infrastructures with an empirical study of user perception and behaviour. By examining spoofing techniques across VoIP systems, SIP trunk interconnections, and mobile networks based on SIM Box infrastructures, the study provides a unified view of how caller identity manipulation is enabled, propagated, and ultimately perceived in real communication scenarios. From a technical perspective, the experimental results demonstrate that caller-ID spoofing feasibility is primarily driven by implicit trust models and validation policies, rather than by protocol vulnerabilities alone. In VoIP environments, spoofing feasibility is largely conditioned by the configuration of SIP servers and PBXs, where permissive policies allow manipulated identities to be accepted at the client, server, or signalling level. SIP trunk interconnections extend this risk across administrative domains, enabling spoofed identities to propagate based on inter-provider trust relationships. In contrast, SIM Box-based spoofing leverages the native trust model of mobile broadband networks, using legitimate mobile identifiers that significantly reduce the effectiveness of traditional signalling-based detection mechanisms. Empirical behavioural evidence further demonstrates that behavioural responses to caller-ID spoofing scenarios are driven primarily by scenario-dependent trust cues and by a clear separation between engagement and compliance processes. While users may answer calls or perceive them as legitimate based on contextual indicators, cybersecurity knowledge and spoofing awareness primarily influence the likelihood of following instructions rather than initial engagement. This confirms that caller-ID spoofing should be understood as a socio-technical phenomenon in which infrastructure trust models and human decision-making jointly shape behavioural responses and perceived credibility. These observations arise from fundamental trust assumptions inherent to modern telephony architectures.

Complementing the technical analysis, the user study reveals that

Table 8
Comparative assessment of defensive mechanisms against Caller-ID spoofing.

Defence Layer	Technology	Mechanism	Tested	10 Attempts	Effectiveness
VoIP	Asterisk Auth	Credential validation	✓	10 blocked	100% mitigation in evaluated scenarios
VoIP	From-Validation	Identity enforcement	✓	10 blocked	100% mitigation in evaluated scenarios
VoIP	CLI Rewriting	Forced consistency	✓	10 rewritten	100% mitigation in evaluated scenarios
Trunk	Provider A	Global routes	✓	10/10 accepted (any CLI, any destination)	0% (no defence)
Trunk	Provider B	Spain-only routes	✓	10/10 domestic accepted; 10/10 international unrouteable	0% for domestic spoofing — routing scope ≠ defence
Trunk	STIR/SHAKEN	Digital signatures	×	—	Not tested — incomplete deployment limits real-world effectiveness

Note: "Blocked" indicates the call was rejected outright; "rewritten" indicates the call proceeded but the spoofed CLI was overwritten with the authenticated identity before reaching the recipient. Both outcomes achieve 100% mitigation of the spoofing attempt's intended effect in the evaluated scenarios (i.e., the recipient never sees the falsified caller ID), though via different mechanisms.

behavioural responses closely mirror infrastructure-level trust assumptions. Survey results show that users rely heavily on contextual trust cues—such as institutional identifiers, number familiarity, and numbering conventions—when assessing call legitimacy. Even among participants with relatively high digital engagement, conceptual awareness of spoofing and smishing remains limited, and many users depend on default protection mechanisms without fully understanding their scope or limitations. Statistical analyses further demonstrate that prior cybersecurity knowledge, perceived protection, and contextual ambiguity interact to shape user behaviour, reinforcing the view that caller-ID spoofing is fundamentally a socio-technical problem. Taken together, the findings indicate that no single countermeasure can effectively address caller-ID spoofing across all telephony environments (Buccafurri et al., 2026). In VoIP and SIP trunk scenarios, mitigation can be significantly improved through stricter validation policies, consistent caller-ID rewriting, and clearly defined trust boundaries between infrastructure components and domains. However, in mobile networks, SIM Box-based spoofing poses a substantially more complex challenge, as it exploits legitimate subscriber identities and the inherent trust of mobile broadband infrastructures. Effective mitigation in this context requires operator-level detection mechanisms based on behavioural analysis, traffic profiling, and strengthened SIM acquisition controls, rather than reliance on identity verification alone. In summary, this work contributes a structured framework for understanding caller-ID spoofing that bridges infrastructure design, trust relationships, and human behaviour. By demonstrating how technical feasibility and user perception jointly influence the credibility and behavioural impact of spoofed identities, the study underscores the need for multi-layered security approaches that combine robust infrastructure-level validation with an informed understanding of user decision-making under uncertainty.

The results therefore suggest that effective mitigation of caller-ID spoofing requires coordinated improvements across both signalling infrastructures and human-centred defences. Strengthening identity validation alone cannot fully address the problem if users continue to rely on inherited trust cues, just as awareness programmes alone cannot compensate for infrastructures that continue to propagate unauthenticated caller identities.

8. Future work

Future research may extend the scope of this work in several complementary directions. Beyond the manipulation and presentation of caller identity itself, an important line of investigation involves the analysis of downstream exploitation scenarios enabled by telephone spoofing. It would be highly relevant to evaluate how caller-ID manipulation can facilitate unauthorized access to services associated with telephone numbering, such as voicemail systems, when these rely on weak authentication mechanisms based on the originating number or on implicit trust assumptions.

In a similar vein, further work should systematically assess the impact of spoofing on auxiliary telephony services, including call forwarding mechanisms, automated notifications, and verification processes based on incoming calls. Understanding to what extent spoofed caller identities can be leveraged as an attack vector in these contexts would provide valuable insight into the cascading effects of identity manipulation across telephony service ecosystems.

Direct evaluation of SIM Box-based spoofing techniques through controlled laboratory implementation would extend the experimental validation beyond IP-based infrastructures to mobile networks. On the technical side, future studies could deepen the evaluation of mitigation mechanisms applicable to VoIP and SIP trunk environments, moving beyond conceptual analysis toward practical deployment scenarios. This includes experimentally assessing the effectiveness of stricter validation policies, controlled caller-ID rewriting, and infrastructure-level monitoring mechanisms under realistic conditions. Such an approach would

allow for a comparative assessment of countermeasures, identifying those that provide meaningful resilience against spoofing attacks while remaining compatible with operational constraints.

Finally, from a methodological perspective, combining infrastructure experiments with longitudinal or adaptive user studies could offer deeper insight into how trust, awareness, and behaviour evolve over time when users are repeatedly exposed to spoofing scenarios. Exploring the interaction between emerging identity-verification frameworks, regulatory measures, and user perception—especially in heterogeneous environments that combine IP-based and legacy networks—remains a promising direction for improving the resilience of modern telephony systems against identity-based threats.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Copilot to translate part of the content of the present manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

CRediT authorship contribution statement

L.A. Oliveira Rodríguez: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis. **I. Benítez Zapico:** Investigation. **D. Melendi Palacio:** Supervision. **R. García Fernández:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.cose.2026.105144](https://doi.org/10.1016/j.cose.2026.105144).

Data availability

Public repository containing configuration files and others may be found at https://github.com/isifire/Caller_ID_Spoofing_Study_UniOvi.

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